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EXPERIMENT 7

SECURITY NETWORK ARCHITECTURE

AIM:

To study and understand the design and implementation of a secure network architecture, including segmentation, firewalls, and access control mechanisms, in TryHackMe platform.

ALGORITHM:

1. Log in to the TryHackMe platform and start the Secure Network Architecture room.
2. Analyze the given network topology, including internal, external, and DMZ segments. ion.
3. Examine the use of firewalls, routers, intrusion detection/prevention systems (IDS/IPS), and access control lists (ACLs).
4. Study how traffic is filtered between different network zones and how rules are applied to allow or deny communication.
5. Verify how segmentation and security controls protect critical systems and reduce attack surfaces.

TASKS:



Secure Network Architecture

Learn about and implement security best practices for network environments.

60 min 46,844

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Room completed (100%)

- Task 1 Introduction
- Task 2 Network Segmentation
- Task 3 Common Secure Network Architecture
- Task 4 Network Security Policies and Controls
- Task 5 Zone-Pair Policies and Filtering
- Task 6 Validating Network Traffic
- Task 7 Addressing Common Attacks
- Task 8 Conclusion

Answer the questions below

How many trunks are present in this configuration?

4

✓ Correct Answer



What is the VLAN tag ID for interface eth12?

30

✓ Correct Answer



Answer the questions below

According to the corresponding ACL policy, will the first packet result in a drop or accept?

accept

✓ Correct Answer

According to the corresponding ACL policy, will the second packet result in a drop or accept?

drop

✓ Correct Answer

Answer the questions below

From the above table, what zone would a user connecting to a public web server be in?

External

✓ Correct Answer



From the above table, what zone would a public web server be in?

DMZ

✓ Correct Answer



From the above table, what zone would a core domain controller be placed in?

Restricted

✓ Correct Answer



Answer the questions below

What is the flag found after filling in all blanks on the static site?

THM{M05tly_53cure}

✓ Correct Answer

Answer the questions below

Does SSL inspection require a man-in-the-middle proxy? (Y/N)

Y

✓ Correct Answer

What platform processes data sent from an SSL proxy?

Unified Threat Management

✓ Correct Answer

Answer the questions below

Where does DHCP snooping store leased IP addresses from untrusted hosts?

DHCP Binding Database

✓ Correct Answer

Will a switch drop or accept a DHCPRELEASE packet?

Drop

✓ Correct Answer

Does dynamic ARP inspection use the DHCP binding database? (Y/N)

Y

✓ Correct Answer

Dynamic ARP inspection will match an IP address and what other packet detail?

MAC Address

✓ Correct Answer

Answer the questions below

Read the above and continue learning!

No answer needed

✓ Correct Answer

RESULT:

All the tasks in secure network architecture tryhackme room have been completed successfully.